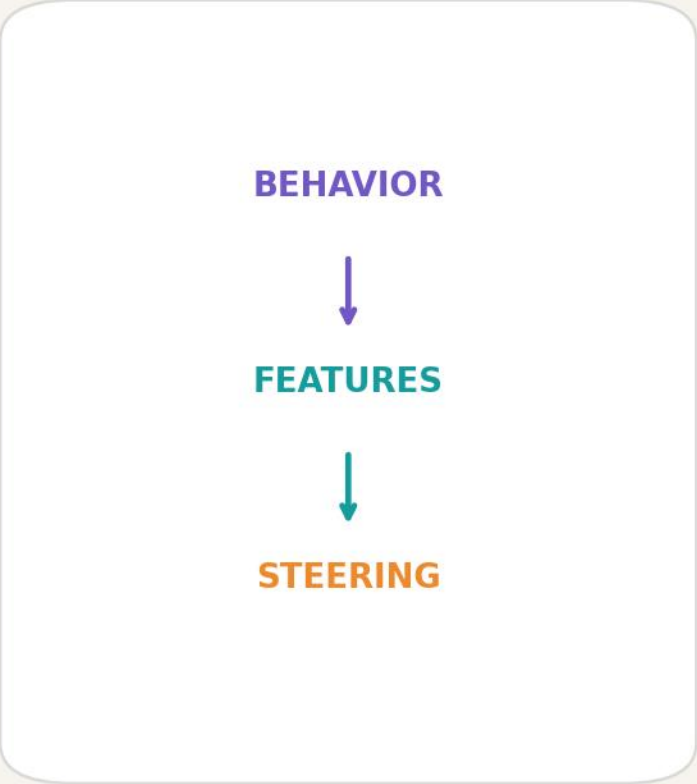
CROSSCODER MODEL DIFFING

FROM MODEL DIFFS TO CAUSAL CONTROL

A shared sparse representation for explaining and testing behavioral differences between code models



RESEARCH QUESTION

Evaluation tells us what changed. CrossCoder asks which internal differences can explain—and control—it.

BEHAVIOR

Which tasks changed?

pass → fail
fail → pass

What failure mode changed?

REPRESENTATION

Paired layer-16 residuals

Same-text alignment

Shared sparse features

CAUSAL TEST

Choose candidate directions

Intervene during generation

Compare against controls

Behavior localizes the model difference. Sparse features turn it into a testable hypothesis.

ONE ANALYSIS FRAMEWORK, TWO VERY DIFFERENT MODEL CHANGE

The same same-text CrossCoder design spans fine-tuning and model merging

DEEPSEEK

specialization through fine-tuning

BASE
layer 16
same evaluated token IDs

→

FINE-TUNED
layer 16
same evaluated token IDs

→

CROSSCODER

16,384 latents · ReLU
exact TopK-100

CODELLAMA

specialization through model merging

BASE
layer 16
same evaluated token IDs

→

MERGED
layer 16
same evaluated token IDs

→

CROSSCODER

16,384 latents · ReLU
exact TopK-100

Different specialization mechanisms; identical analytical question: which shared latent differences track—and control—behavior?

THE TWO SPECIALIZATIONS MOVE PERFORMANCE IN OPPOSITE DIRE

Canonical extraction-v4 labels · same raw generations, audited reprocessing

DEEPSEEK · BIGCODEBENCH

268/1140
BASE

404/1140
FINE-TUNED

+11.93 pp
BigCodeBench pass-rate change

CODELLAMA · BIGCODEBENCH

314/1140
BASE

27/1140
MERGED

-25.18 pp
BigCodeBench pass-rate change

DeepSeek: 215 improvements · 79 regressions

CodeLlama: 4 improvements · 291 regressions

RECATCHER CONTEXT

Directionally consistent motivation: specialization can improve aggregate capability while still creating localized regressions.

Not a direct numerical replication: ReCatcher used 10 generations/prompt and different regression definitions.

WE FOCUS ON TWO ONE-SIDED BEHAVIORAL TRANSITIONS

Taxonomy turns aggregate model differences into semantically targeted cohorts

DEEPSEEK · 215 IMPROVEMENTS

base fail → fine-tuned pass

119/215: generated test/import contamination
48: wrong output or logic
16: missing name/import
32: other primary categories

Mechanistic focus: 80 contamination cases

CODELLAMA · 291 REGRESSIONS

base pass → merged fail

120/291: API/type mismatch
50: wrong logic/other runtime
43: edge case/exception
78: generation, import, syntax, commentary

Mechanistic focus: 50 wrong-logic/runtime cases
Improvements: only 4 positives · EXCLUDED

HOW WE CLASSIFIED

Rule/AI-assisted taxonomy + human spot checks

IMPORTANT QUALIFICATION

Primary = most salient observed error—not a unique root cause. CodeLlama failures are heterogeneous and oft

DEEPSEEK /5 · CONTAMINATION

BASE · FAIL
Correct-looking body continues into:
`#tests/test_task_func.py`
`import pytest`

FINE-TUNED · PASS
Completes the function and stops after a short example.

First textual divergence ≈76% of shorter code;
target boundary is late/post-solution.

CODELLAMA /490 · WRONG LOGIC/API CONTRACT

BASE · PASS
parses XML → writes JSON file → returns result

MERGED · FAIL
`return xmltodict.parse(s)`
omits the required file write side effect

First divergence ≈81% of base code;
target region is the late function body.

FEATURE SCREENING

We rank stable behavioral contrasts—not raw activation alone

4 SUMMARIES

max · early_max
mean · active fraction

→

Δ PER TASK

aggregate(specialized text)
− aggregate(base text)

→

OBSERVED EFFECT

mean Δ positives
− mean Δ controls

→

200 PERMUTATIONS

shuffle labels
recompute effect

→

E/V

effect ÷ null SD

Minimum support

max(3 tasks, 10% of positives)

Interpretation

E/V is permutation signal-to-noise—not a z-score or p-value.

Positive and negative orientations are retained: either enrichment direction may support behavior-aligned steering.

EIGHT COMPLEMENTARY SCREENING CELLS PER MODEL

Transition type × comparison design × temporal scope

DEEPSEEK

REGRESSION

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

IMPROVEMENT

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

CODELLAMA

REGRESSION

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

IMPROVEMENT

ONLY 4 POSITIVES · EXCLUDED

ASSOCIATION

global · local

PAIRED MODEL Δ

global · local

Local = ±10% around the first normalized divergence—not a semantic error category.

WHAT THE SCREENING ACTUALLY RETURNS

Representative valid cells; E/V is the primary ranking signal

DEEPSEEK · IMPROVEMENT ASSOCIATION

Rank	Feature	Effect	E/V
1	3602	-0.021	**-8.23**
2	10168	-0.018	**-7.67**
3	5422	-0.005	**-7.66**
4	1982	-0.015	**-7.53**
5	1862	-0.008	**-7.45**
6	1650	-0.026	**-7.28**
7	5039	-0.088	**-7.26**
8	9580	-0.013	**-7.24**
9	5837	-0.010	**-7.18**
10	9537	-0.038	**-7.15**

CODELLAMA · REGRESSION ASSOCIATION

Rank	Feature	Effect	E/V
1	13147	+0.126	**+5.24**
2	8992	-0.011	**-4.93**
3	1840	-0.046	**-4.77**
4	16161	+0.127	**+4.76**
5	15471	+0.005	**+4.75**
6	10570	+0.188	**+4.60**
7	8138	-4.973	**-4.51**
8	7353	+0.228	**+4.48**
9	14359	-0.166	**-4.46**
10	2825	+0.338	**+4.38**

Examples—not a single global ranking. Candidate pools are formed by union and deduplication across valid cells.

ERROR-FOCUSED TOP-10 LISTS DEFINE THE NEW α=3 CANDIDATE POOL

Three summaries per cell: max · mean · active fraction; ranking by absolute E/V

DEEPSEEK · 80 CONTAMINATION IMPROVEMENTS

ASSOC · GLOBAL	ASSOC · LOCAL	PAIRED · GLOBAL	PAIRED · LOCAL
13879	14175	**2913**	2468
6895	9666	**7728**	14295
1078	1206	15821	8661
596	8686	12308	**7728**
8185	7169	16264	**2913**
14820	11612	**2957**	**2957**
1939	7804	**3602**	9700
15235	**1939**	2621	6305
15733	4994	6877	10736
16205	1591	14006	**3602**

40 nominations → 35 unique features

5 feature(s) appear in more than one list

CODELLAMA · 50 WRONG-LOGIC REGRESSIONS

ASSOC · GLOBAL	ASSOC · LOCAL	PAIRED · GLOBAL	PAIRED · LOCAL
7692	**15559**	8875	**4173**
10818	3097	**7588**	**15559**
5642	565	12895	12623
447	**4173**	**447**	15968
11596	15890	**4634**	8142
1017	**414**	5862	9419
4634	4740	3366	13231
7588	13768	7486	**414**
3442	4309	**1017**	13151
8384	14969	**5642**	13172

40 nominations → 32 unique features

8 feature(s) appear in more than one list

NEW α=3 SWEEP: exactly 35 DeepSeek + 32 CodeLlama unique features · no post-hoc additions

Hard gate: baseline must reproduce the original raw completion byte for byte before any steered arm runs.

SCREEN FIRST; TEST CAUSALLY AT A COMMON OPERATING POINT

The statistical shortlist reduces the intervention search by ~99.8%

16,384 → **≈30–40** → **≈0.2%**

latents / CrossCoder

candidates / model

of latent space

WHY |α| = 3?

Earlier dose-response work suggested a practical exploratory point: visible causal changes below the most disruptive high-dose regime.

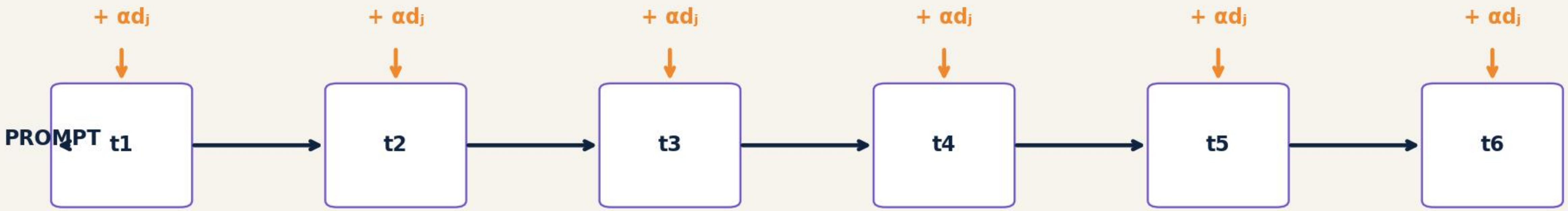
WHY STANDARDIZE?

One magnitude makes dozens of candidates comparable before full dose curves, reverse-model t

|α|=3 is an operating point—not an independently validated optimum.

OUR CURRENT INTERVENTION IS CONTINUOUS STEERING

The selected decoder direction is added at every autoregressive generation step



CURRENT PROTOCOL

Layer 16 · last residual position
feature decoder on intervened model side
applied through the full generation

WHAT IT IS NOT

Not a one-token intervention
not activation-gated or clamped
not conditional on natural activation

This tests continuous control of the generation trajectory—not whether one naturally active token alone caused the error.

THE COMPLETE RESULT SEPARATES A STRONGER AND A WEAKER REGIM

Official BigCodeBench 0.1.5 evaluation · α=3 · continuous last-token steering

DEEPSEEK

Features evaluated	35/35
Baseline	0/80
Feature-task transitions	116
Unique corrected tasks	16/80

Several features produce repeated official corrections.

CODELLAMA

Features evaluated	32/32
Baseline	0/50
Feature-task transitions	6
Unique corrected tasks	4/50

Single-feature corrections remain sparse.

STRONGER / LOCALIZED

DeepSeek contains several multi-task causal candidates.

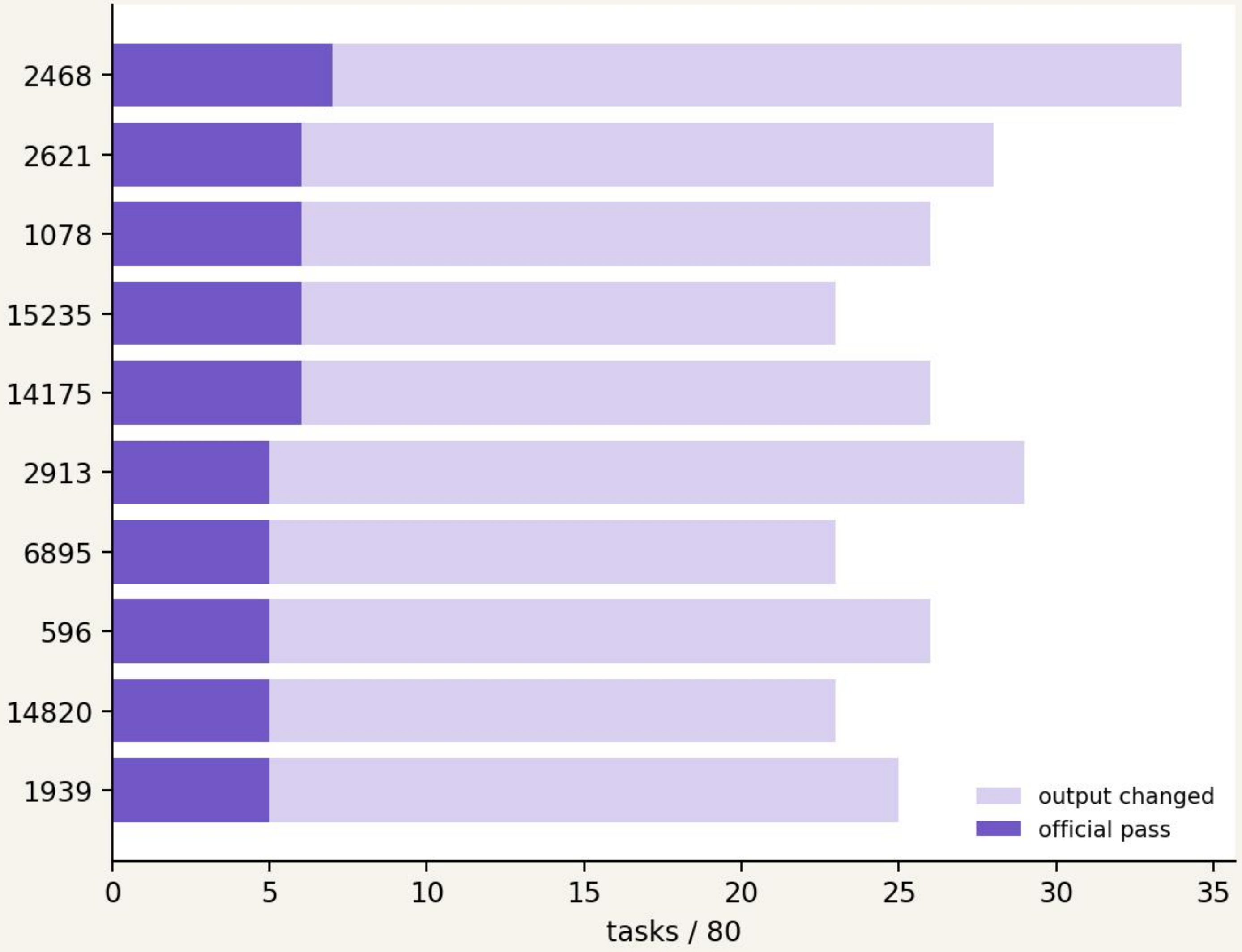
WEAKER / DIFFUSE

CodeLlama yields sparse, task-specific corrections.

COMPLETE · DeepSeek 35/35 and CodeLlama 32/32; canonical α=0 gates passed before steering.

DEEPSEEK: THE COMPLETE α=3 RANKING ADDS TWO TOP-5 FEATURES

Causal outcome first; |E/V| breaks ties among features with equal official passes



FINAL TOP 5

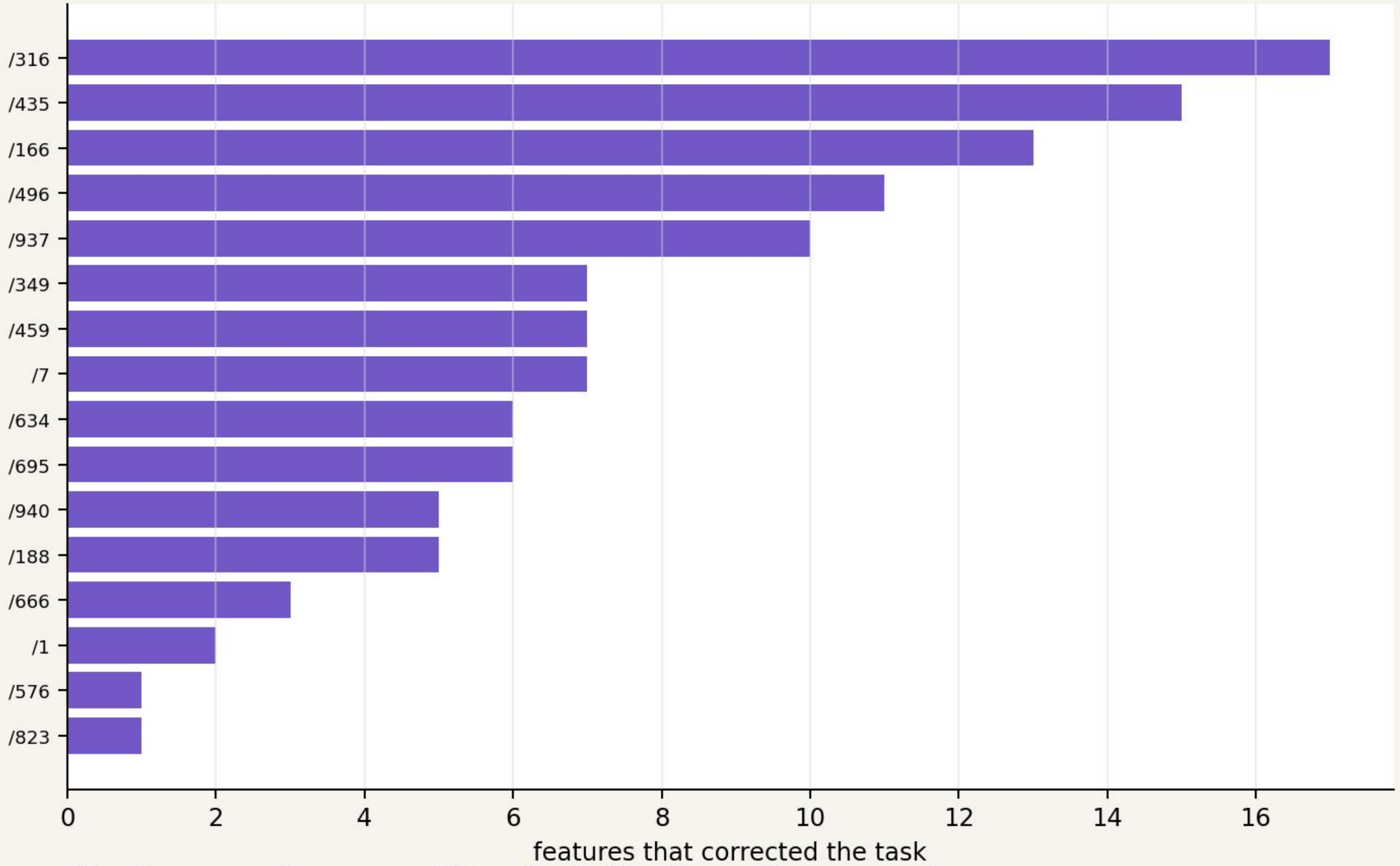
1. 2468   7/80 pass   34/80 changed   E/V -7.29
2. 2621   6/80 pass   28/80 changed   E/V -10.53
3. 1078   6/80 pass   26/80 changed   E/V +5.72
4. 15235   6/80 pass   23/80 changed   E/V -4.61
5. 14175   6/80 pass   26/80 changed   E/V +4.09

1078 and 14175 enter after completion of all 35 arms.

2913 is #6 at α=3, but later peaks at 11/80 at α=−5.

DEEPSEEK: 116 TRANSITIONS COLLAPSE TO 16 SUSCEPTIBLE TASKS

Feature-task outcomes are not independent causal successes



16 unique corrections across 80 contamination-focused improvements.

CONCENTRATION

/316 17 features
/435 15 features
/166 13 features
/496 11 features
/937 10 features

INTERPRETATION

The sweep finds causal levers, but a small set of tasks is broadly steering-

CANONICAL ERROR-FOCUSED α=3 SWEEP · PARTIAL READOUT

FEATURE 2468 IS THE LEADING DEEPSEEK CANDIDATE

Meaning ↔ failure mode ↔ activation timing must remain distinct

SCREENING

paired local · rank 1
E/V -7.29
summary: max
selected around the late divergence window

CAUSAL OUTCOME

7/80 official passes
34/80 outputs changed
13/80 contamination cleanups
baseline: 0/80

CAUSED TIMING

Among changed outputs:
mean first divergence: 42.7%
median: 40.8%

Continuous steering can redirect earlier than the natural activation windo

SUPPORTED local screening nominated a feature with repeated official causal effects.

NOT YET SUPPORTED feature-specificity versus sham/matched latent controls.

CODELLAMA: SINGLE-FEATURE CONTROL IS WEAK AND TASK-SPECIFIC

Complete 32-feature sweep; baseline 0/50

SIX FEATURES · ONE PASS EACH

7692 → BigCodeBench/119
10818 → BigCodeBench/119
5642 → BigCodeBench/630
11596 → BigCodeBench/119
4309 → BigCodeBench/490
8142 → BigCodeBench/823

FOUR UNIQUE TASKS

/119 corrected by 3 features
/823 corrected by 1
/490 corrected by 1
/630 corrected by 1

26/32 features produced no official correction.

Trajectory changes occur earlier and more diffusely than in the contamination-focused DeepSeek cohort.

Boundary condition: correlation and trajectory control do not imply that one latent can repair a heterogeneous merged-model regression.

CANONICAL ERROR-FOCUSED α=3 SWEEP · PARTIAL READOUT

NEXT TARGETS COMBINE CAUSAL EFFECT WITH SCREENING STRENGTH

Ordering: official passes first; |E/V| breaks ties

DEEPSEEK

RANK	FEATURE	PASS	BEST SCREEN	|E/V|
1	**2468**	**7/80**	**paired local · #1**	**7.29**
2	2621	6/80	paired global · #8	10.53
3	15235	6/80	association global · #8	4.61
4	14175	6/80	association local · #1	4.09
5	2913	5/80	paired global · #1	10.95

CODELLAMA

RANK	FEATURE	PASS	BEST SCREEN	|E/V|
1	**7692**	**1/50**	**association global · #1**	**5.90**
2	10818	1/50	association global · #2	5.90
3	5642	1/50	association global · #3	5.74
4	11596	1/50	association global · #5	5.61
5	4309	1/50	association local · #9	4.86

These are candidates for full dose curves plus sham and randomly selected/matched latent controls—not confirmatory winners yet.

THE α=3 SWEEP REVEALS BOTH MODEL AND TASK SENSITIVITY

Error-focused candidates · continuous last-token steering · official BigCodeBench 0.1.5

DEEPSEEK · CONTAMINATION IMPROVEMENTS

25/35 features evaluated
83 feature-task transitions
14/80 unique tasks corrected

Several features correct multiple tasks.

CODELLAMA · LOGIC/RUNTIME REGRESSIONS

30/32 features evaluated
6 feature-task transitions
4/50 unique tasks corrected

Single-feature corrections are sparse.

READING

The sweep reduces 16,384 latents to causal candidates, but repeated correction of the same task also reveals task susceptibility. Dose curves and controls are required next.

CAUSAL STEERING · CANONICAL SEED

FROM α=3 SCREENING TO CONTROLLED DOSE–RESPONSE

The confirmatory design separates trajectory control, outcome control, directionality, and specificity

1 α=0 GATE

byte-exact canonical reproduction

2 TARGET CURVE

|α| = 1 · 2 · 3 · 4 · 5

3 REVERSE TEST

opposite model and direction

4 CONTROLS

3 random latents + 3 orthogonal shams

5 TWO ENDPOINTS

evaluated code change + official pass

INTERVENTION

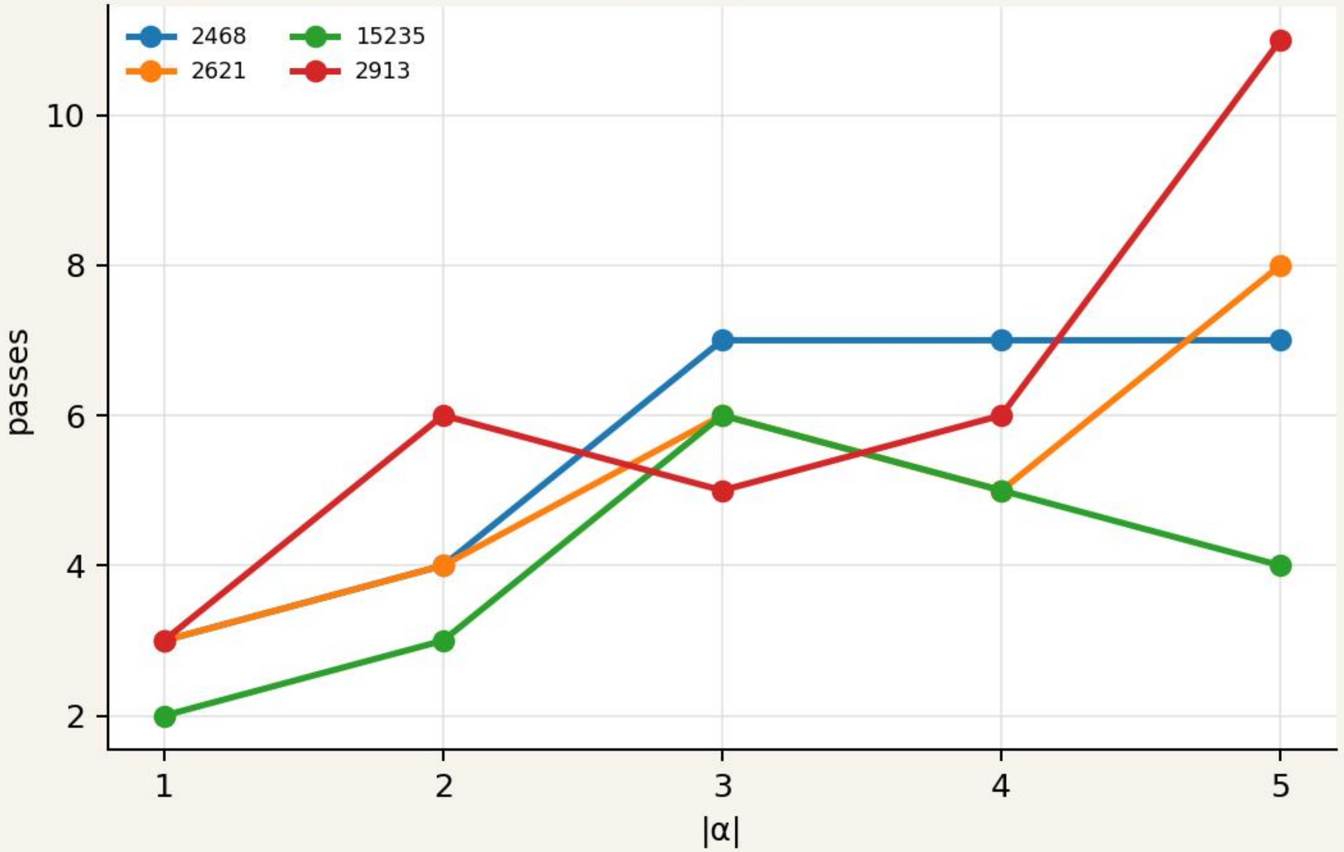
Layer 16 · selected decoder side · last residual position · added at every autoregressive generation step · temperature 0.2 · top-p 0.95 · max 512 · seed = 1000 + task_idx × 100

α=0 passed: DeepSeek 80/80 byte-exact · CodeLlama 50/50 byte-exact.

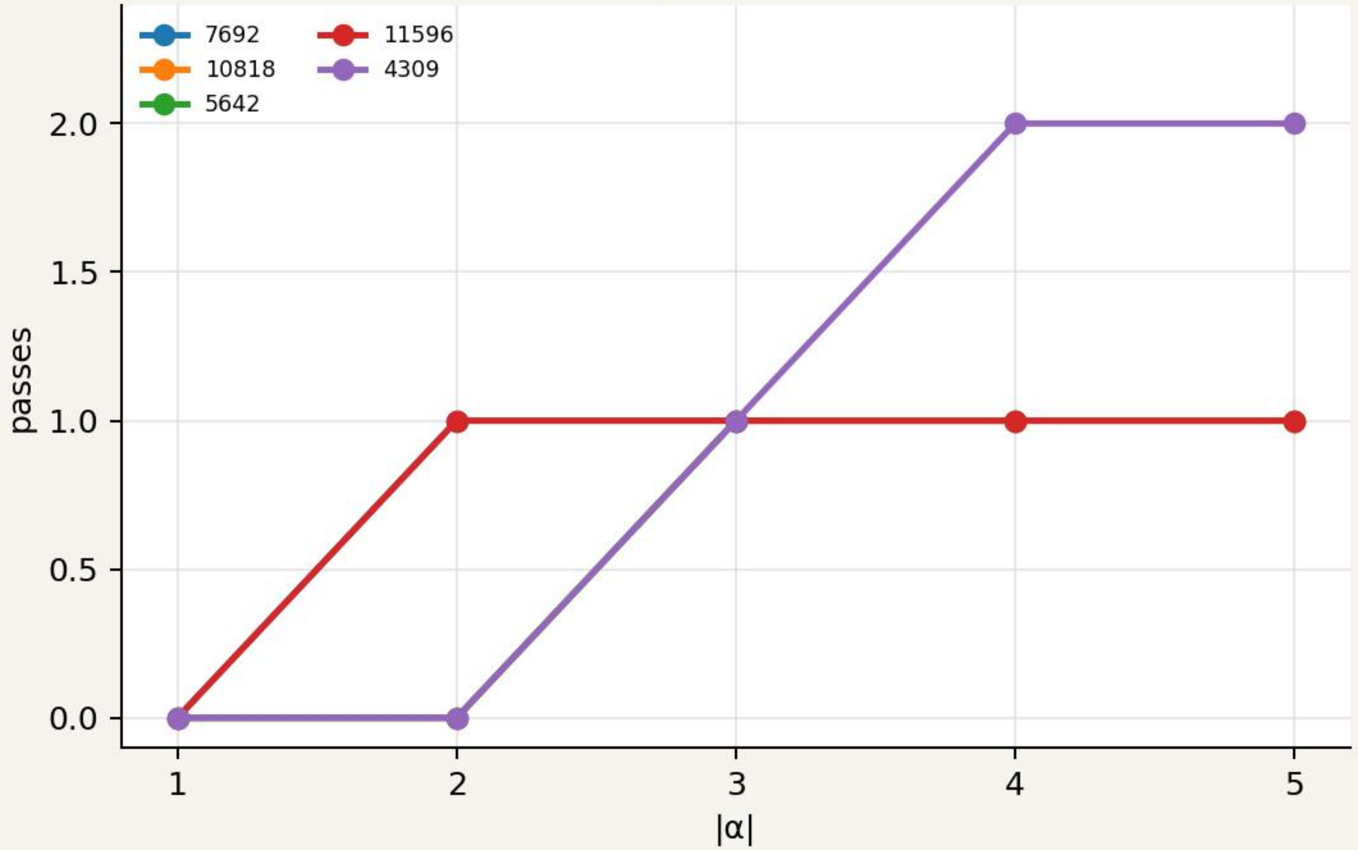
STEERING CHANGES TRAJECTORIES—AND DEEPSEEK OUTCOMES

Complete direct target curves; α=3 integrated from the canonical sweep

DeepSeek · official fail→pass / 80



CodeLlama · official fail→pass / 50



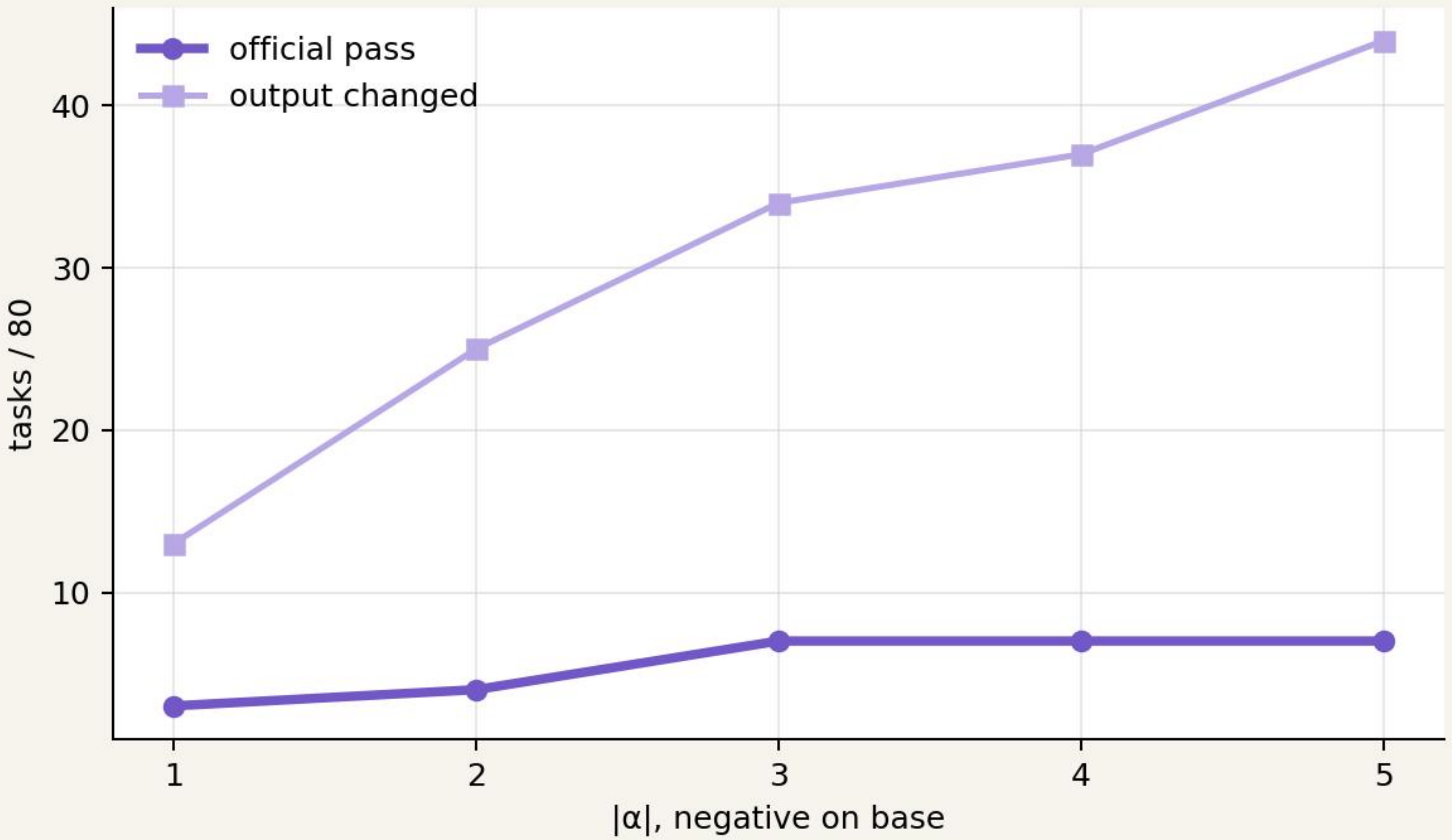
DEEPSEEK 16 unique tasks corrected; peak 11/80.

CODELLAMA 5 primary tasks corrected; peak 2/50.

Specificity versus the new random/sham controls remains pending; causal trajectory and outcome effects are already directly measured.

FEATURE 2468: THE LEADING DEEPSEEK MECHANISTIC CANDIDATE

Screening strength and stable causal response align around the contamination boundary



SCREENING

paired local · rank #1
base-enriched · max
E/V = −7.29
support: 62/80

CAUSAL READOUT

7/80 passes at α=−3, −4 and −5
34 → 44 outputs changed

Meaning ↔ failure mode ↔ activation timing

Interpretation: a stable lever over late post-solution behavior.

CAUSAL STEERING · CANONICAL SEED

DEEPSEEK /7: STEERING REMOVES POST-SOLUTION TEST CONTAMINATION

Feature 2468 · α=−4 · official fail→pass

BASE · FAIL

```
return top_product

# test_task.py
import unittest
from task import task_func

class TestTask(unittest.TestCase):
    ...
```

STEERED · PASS

```
return top_product

# task_func("path/to/sales.csv")
# task_func("sales.csv")

[no executable test harness]
```

MECHANISTIC READING

The useful function is preserved; the principal intervention occurs after the solution boundary and removes executable test/import contamination. This is a clean outcome-level example, not proof that every 2468 corr

FEATURE 2468 ALSO MAKES COHERENT CHANGES THAT DO NOT REPAIR TH

Semantic control is not equivalent to sufficient correction

/97 · STILL FAILS

Only expected floating-point constants inside generated tests change.

Trajectory changed late, but the irrelevant test harness remains.

/823 · STILL FAILS

Generated test precision and comments change near the end.

The intervention affects post-solution material without repairing the evaluated function.

WHY THIS MATTERS

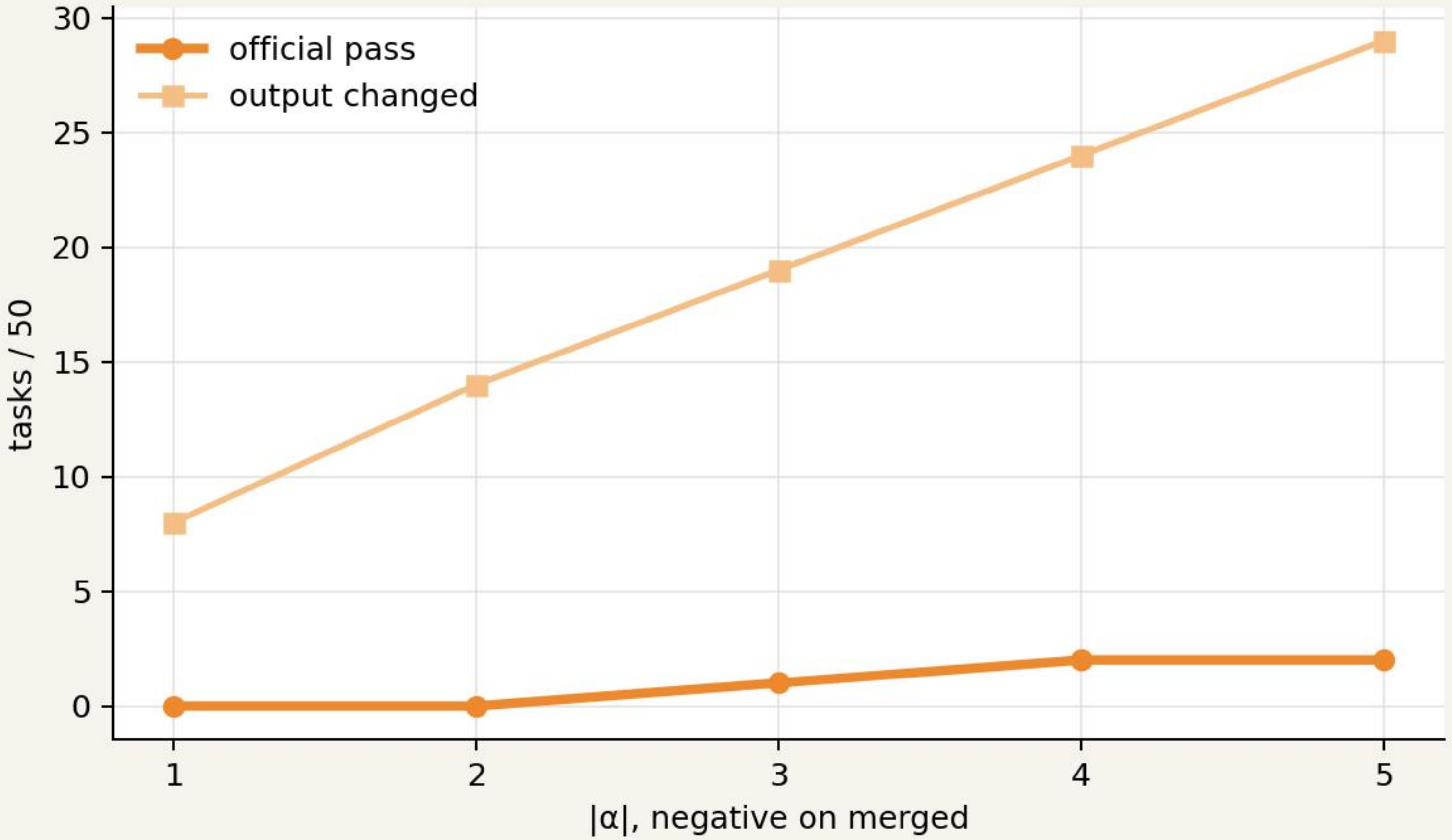
The feature is behaviorally coherent, but cleanup is neither universal no

Observed: directional control over late generated content.

Not implied: a single-feature explanation for all 80 failures.

FEATURE 4309: CODELLAMA’S STRONGEST CURRENT OUTCOME CANDIDAT

Trajectory effects grow with dose; official repairs remain rare



OUTCOME

0 · 0 · 1 · 2 · 2 passes
8 → 29 outputs changed
primary successes include /183 and /490

BOUNDARY CONDITION

More trajectory changes do not translate proportionally into correct programs.

Current interpretation: distributed regression mechanisms.

CAUSAL STEERING · CANONICAL SEED

CODELLAMA /490: STEERING REPAIRS PARSING, WRITING, AND RETURN

Feature 4309 · α=−5 · official fail→pass

MERGED · FAIL

```
return json.loads(s)

[does not parse XML correctly]
[does not write the requested JSON]
[one generated function returns nothing]
```

STEERED · PASS

```
result = xmltodict.parse(s)
with open(file_path, "w") as f:
    json.dump(result, f)
return result
```

MECHANISTIC READING

The intervention produces a semantically aligned multi-part repair. Because CodeLlama successes are sparse, this is evidence for selected-task causal control—not a broad single-feature account of merged-model reg

CODELLAMA CHANGES CODE FREQUENTLY WITHOUT REPAIRING IT

Feature 4309 exposes trajectory control and limited outcome specificity

/738 · STILL FAILS

The intervention removes a closing parenthesis in multiple doctest examples.

The code changes consistently, but in a harmful or irrelevant way.

DOSE RESPONSE

Outputs changed:
8 · 14 · 19 · 24 · 29

Official passes:
0 · 0 · 1 · 2 · 2

INTERPRETATION

The latent can move the trajectory, but the 50-task logic/runtime cohort

Causal effect: supported.

Broad repair mechanism: not supported.

SYNTHESIS

THE EVIDENCE CHAIN MAKES EACH CLAIM AUDITABLE

Association, interpretation, causality, and specificity remain separate layers

TRANSITION → ERROR-FOCUSED SCREEN → TOKENS + TIMING → DIRECTION → α=0 GATE → DOSE CURVE → SEMANTIC CHANGE → FAIL→PASS → CONTROLS

ASSOCIATIVE → INTERPRETATIVE

transition · screening · tokens/timing

CAUSAL → SPECIFICITY

gate · intervention · code/outcome · controls

SYNTHESIS

WHAT THE CURRENT RESULTS SUPPORT—AND WHAT THEY DO NOT

Claims are tiered by evidential strength

SUPPORTED NOW

- screening compresses 16,384 latents
- $\alpha=0$ reproduces canonical generations
- selected features causally alter code
- some interventions yield repeated official passes
- DeepSeek has stronger single-feature outcome control
- CodeLlama is a useful boundary condition

NOT YET SUPPORTED

- every transition is feature-caused
- every target beats new random/sham controls
- one feature explains heterogeneous CodeLlama errors
- 14175 has a complete predicted-direction curve
- generalization across new seeds/tasks

Specificity is a result to be demonstrated—not inferred from screening or one successful task.

SYNTHESIS

A REUSABLE METHODOLOGY FOR CAUSAL MODEL DIFFING

The output is a falsifiable mechanism hypothesis—not merely a feature ranking

1

Paired behavioral transitions

2

Error taxonomy + focused cohort

3

Same-text sparse representation

4

Conditioned E/V screening

5

Meaning + timing + failure mode

6

Directional hypothesis

7

Canonical α=0 gate

8

Dose + reverse steering

9

Official outcome + controls

Every stage can reject, refine, or narrow the hypothesis before a broad mechanistic claim is made.

MODEL DIFFERENCES CAN BECOME TESTABLE CAUSAL HYPOTHESES

A shared representation connects behavioral contrast to controlled intervention

Behavioral differences → **shared sparse features** →

interpretable timing + semantics → **causal steering**

SUPPORTED CONCLUSION

Shared CrossCoder features can localize model differences, generate directional causal hypotheses, and—in selected cases—produce reproducible behavioral transitions.

DeepSeek: more localized outcome control · CodeLlama: causal trajectory control, but more distributed failures.